

Utah Crash Analysis (2015–2019)

Final Capstone Project

Lizbeth Pacheco | Data Technology

About the Data

Data Source

- Utah Open Data Portal
- Public crash records from 2015
2019
- Data collected by Utah agencies

Dataset Overview

- 263,851 crash records
- Includes crash dates, location,
county, and city
- Contains driver risk factors
 - DUI
 - Distracted
 - Drowsy
 - Teen Driver

Limitations

- 2019 contains partial data
- Analysis is only based off on
reported accidents

Key Terms

- Crash Severity – severity levels
from 1 – 5
- Unrestrained – if occupants
were wearing seatbelts

Problem Statement & Research Questions

Problem Being Investigated

This project looks at where crashes happen, when they happen, and what factors may be involved.

Why It Matters

This project looks at where crashes happen, when they happen, and what factors may be involved.

Motivation

Since traffic safety is something that affects everyday drivers, the topic felt meaningful and relevant.

Research Questions

- How did crash totals change from 2015 – 2019?
- Which Utah county had the highest number of crashes?
- Which cities saw the highest number of crashes?
- Which factors appeared the most in the crash dataset?

Data Preparation

Data Cleaning

- Imported the raw dataset from **Google BigQuery** to **Google Colab**
- Reviewed missing values and overall data quality
- Performed exploratory data analysis (EDA)

Data Processing

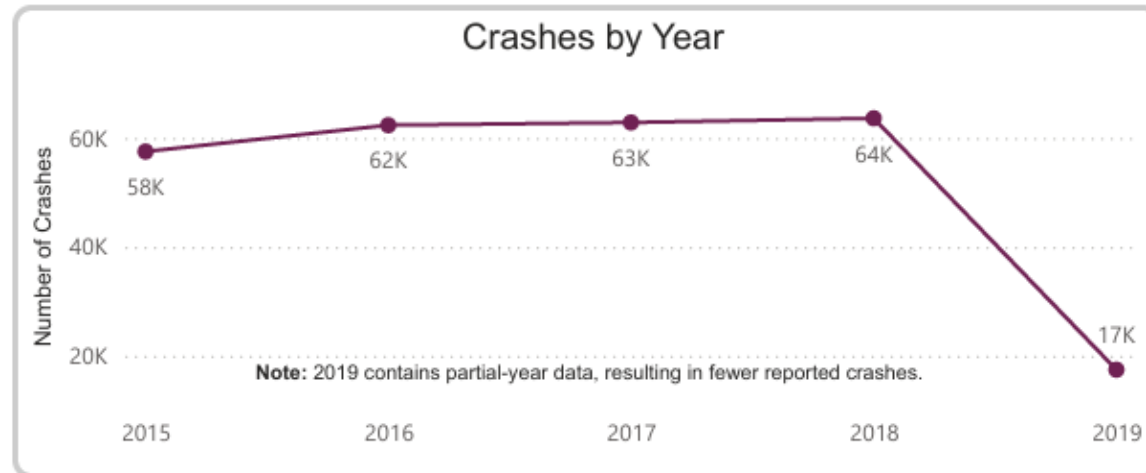
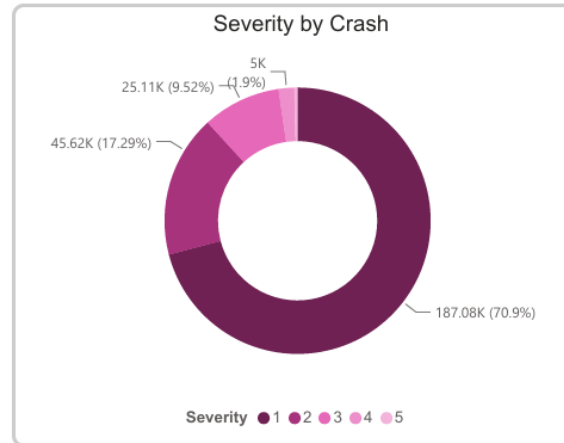
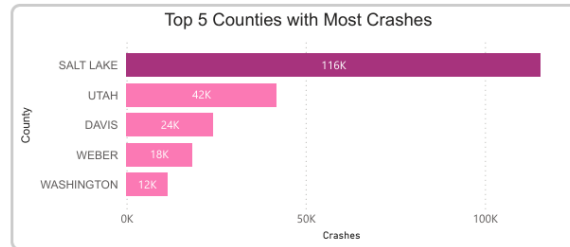
- Uploaded the clean dataset in **Google BigQuery**
- Connected **Google BigQuery** to **Power BI**

Dashboard Development

- Developed three dashboards
- Created DAX measures and calculated columns



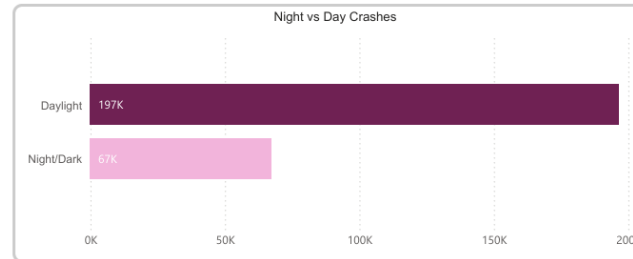
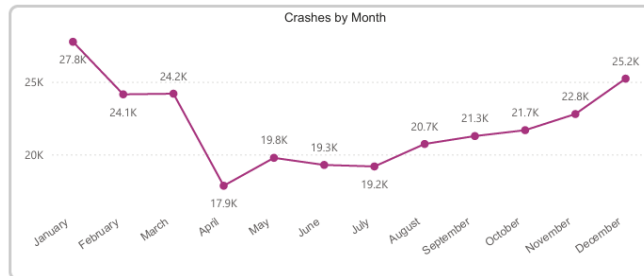
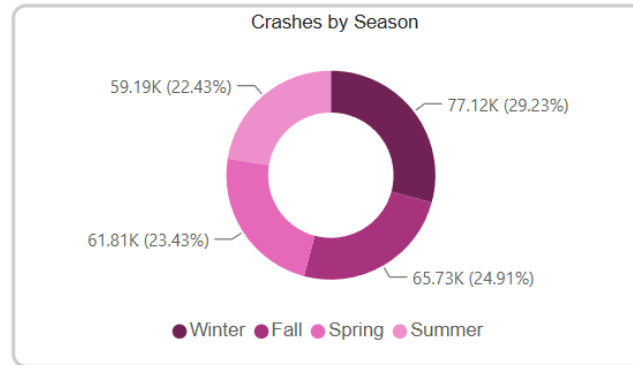
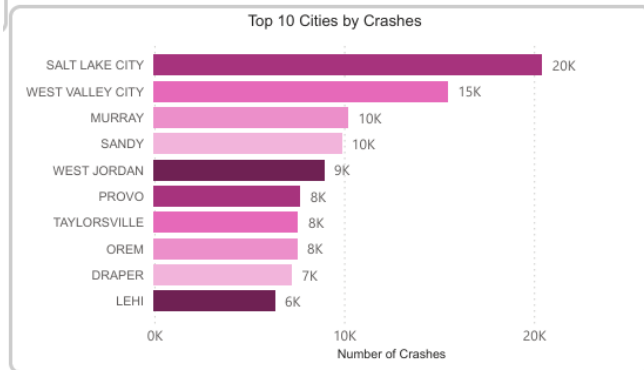
Utah Crash Overview



Key Findings

- Most crashes were classified as Severity Level 1
- Salt Lake County had the most reported crashes
- Crash totals see an increase from 2015 – 2018
 - 2019 only contains partial data, resulting in fewer record

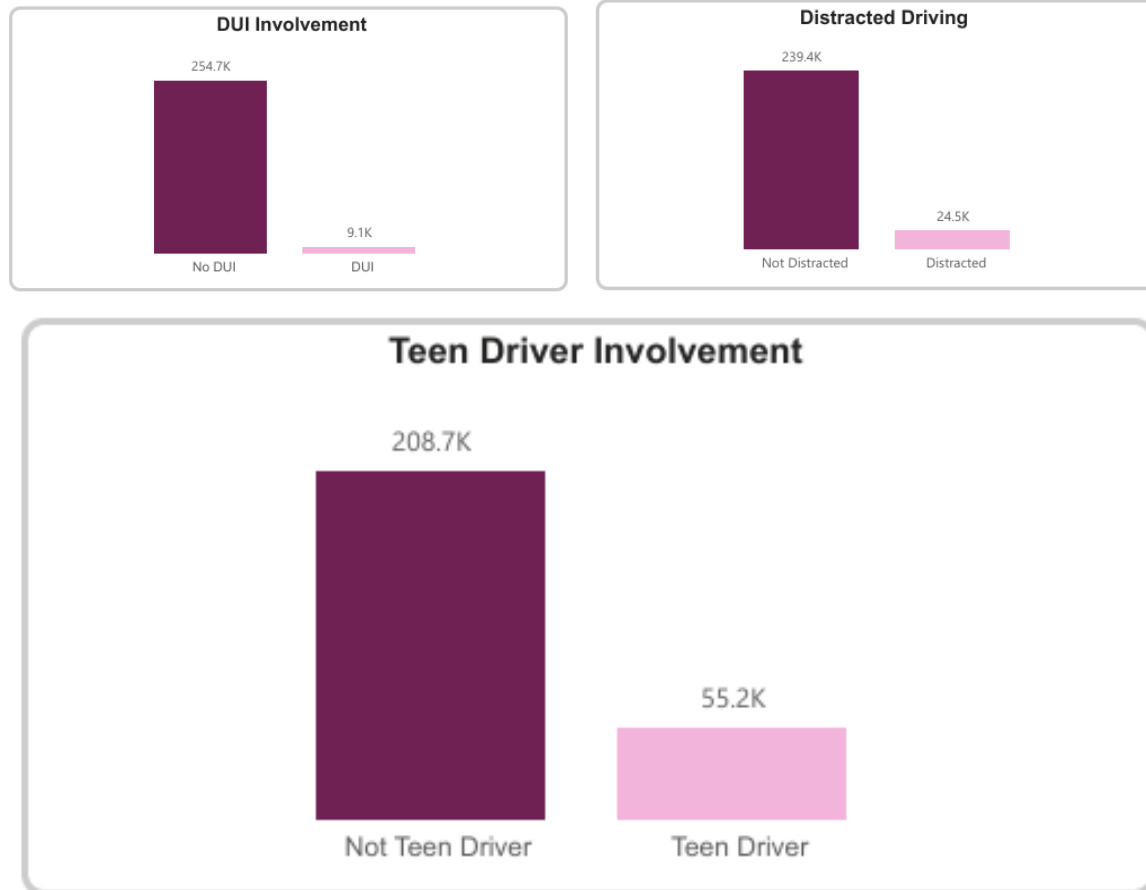
Utah Crash Trends by Time & Location



Key Findings

- January had the highest number of crashes reported
- Daytime crashes occurred more often than nighttime crashes
- Salt Lake County cities have the highest number of crashes
- The winter season had the most reported crashes

Utah Crash Contributing Factors



Key Findings

- Most crashes did not include the selected risks.
- Teen driver accidents were seen more than distracted or DUI crashes
- DUI crashes made up a small portion of all the records

Conclusion and Recommendations

Major Findings

- Salt Lake County reported the highest number of crashes
- Crash totals remained consistent through 2015 to 2018
- Seasons with colder months had the most recorded number of crashes
- Teen driver involvement was more common than DUI or distracted driving among the selected risk factors.

Recommendations

- Continue traffic safety efforts in high-crash counties and cities.
- Increase education and awareness for teen drivers.
- Focus on reducing distracted and impaired driving through public safety campaigns.